

Naive Bayes DNA Heatmap | GainzAlgo

```
//@version=6
indicator("Naive Bayes DNA Heatmap | GainzAlgo", overlay=false)

// -----
// Part 1 defining the UI
// -----
pal_type      = input.string("Traditional", "DNA Palette", options=["Traditional", "Viridis", "Plasma"])
use_data_mode = input.string("Footprint", "Analysis Mode", ["1-Minute", "Footprint"])
pi           = input.bool(true, "Plot Power Index")
nb_train      = input.int(500, "NB Train Lookback")

// Neon Aesthetics
pal_elite_bull = color.rgb(0, 255, 204)
pal_elite_bear = color.rgb(255, 0, 255)
pal_neon_gold  = color.rgb(255, 215, 0)
bg_neutral     = #1a1a1a

get_dna_color(float prob, string type) =>
    float p = nz(prob, 50.0)
    color c = bg_neutral
    if type == "Traditional"
        c := p < 20 ? #ff0055 : p < 40 ? #aa0033 : p < 60 ? bg_neutral : p < 80 ? #00aa44 : #00ff66
    else if type == "Viridis"
        c := p < 20 ? #440154 : p < 40 ? #3b528b : p < 60 ? #21918c : p < 80 ? #5ec962 : #fde725
    else if type == "Plasma"
        c := p < 20 ? #0d0887 : p < 40 ? #7e03a8 : p < 60 ? #bd3786 : p < 80 ? #ed7953 : #f0f921
    c

// Global Logic Variables
var float main_prob = 0.5
var int main_sig = 0
var float heatmap_source = 50.0
var float secondary_metric = 0.0
var float fp_buy_vol = 0.0
var float fp_sell_vol = 0.0
```

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// -----
// part 2 creating the functions for analysis
// -----
f_pdf(x, mean, variance) =>
    float v = math×max(variance, 0.000001)
    (1 / math.sqrt(2 * math.pi * v)) * math.exp(-math.pow(x - mean, 2) / (2 * v))

f_naive_bayes(float feat1, float feat2, float target, int len) =>
    m1_f1 = ta.sma(target > 0 ? feat1 : na, len), m1_f2 = ta.sma(target > 0 ? feat2 : na, len)
    m0_f1 = ta.sma(target <= 0 ? feat1 : na, len), m0_f2 = ta.sma(target <= 0 ? feat2 : na, len)
    v1_f1 = math.pow(ta.stdev(target > 0 ? feat1 : na, len), 2), v1_f2 = math.pow(ta.stdev(target > 0 ?
feat2 : na, len), 2)
    v0_f1 = math.pow(ta.stdev(target <= 0 ? feat1 : na, len), 2), v0_f2 = math.pow(ta.stdev(target <=
0 ? feat2 : na, len), 2)
    p1 = nz(ta.sma(target > 0 ? 1.0 : 0.0, len), 0.5)
    l1 = f_pdf(feat1, nz(m1_f1), nz(v1_f1)) * f_pdf(feat2, nz(m1_f2), nz(v1_f2)) * p1
    l0 = f_pdf(feat1, nz(m0_f1), nz(v0_f1)) * f_pdf(feat2, nz(m0_f2), nz(v0_f2)) * (1.0 - p1)
    prob = nz(l1 / (l1 + l0 + 0.000001), 0.5)
    [prob, prob > 0.5 ? 1 : 0]

if use_data_mode == "Footprint"
    footprint reqFP = request×footprint(100, 70)
    if not na(reqFP)
        fp_buy_vol := reqFP.buy_volume()
        fp_sell_vol := reqFP.sell_volume()
        fp_delta = reqFP.delta()
        volume_row pocRow = reqFP×poc()
        fp_poc_dist = (close - ((pocRow.up_price() + pocRow.down_price()) / 2)) / close * 100
        [prob, sig] = f_naive_bayes(fp_delta, fp_poc_dist, (close > close[1] ? 1 : 0), nb_train)
        main_prob := prob, main_sig := sig, secondary_metric := fp_delta
        heatmap_source := math.max(0, math.min(100, (fp_delta / ta.sma(math.abs(fp_delta + 0.0001),
100) * 50) + 50))
    else
        float[] v_arr = request.security_lower_tf(syminfo.tickerid, "1", volume)
        float[] c_arr = request.security_lower_tf(syminfo.tickerid, "1", close)
        float[] o_arr = request.security_lower_tf(syminfo.tickerid, "1", open)
        float sum_rel_vol = 0.0, float sum_vol_force = 0.0

```

```

int sz = array.size(v_arr)
if sz > 0
  for i = 0 to sz - 1
    v = array.get(v_arr, i), c = array.get(c_arr, i), o = array.get(o_arr, i)
    sum_rel_vol += v / nz(ta.sma(volume, 50), 1.0)
    sum_vol_force += ((c - o) / math.max(v, 1)) * 10000
  secondary_metric := sum_rel_vol / sz
  [prob, sig] = f_naive_bayes(secondary_metric, (sum_vol_force / sz), (close > close[1] ? 1 : 0),
nb_train)
  main_prob := prob, main_sig := sig
  heatmap_source := nz(ta.sma(close > close[1] ? 1.0 : 0.0, 14) * 100, 50.0)

// -----
// Part 4 plotting the heatmap
// -----
p(len) => nz(ta.sma(heatmap_source, len), 50.0)

plot(1, "L1", color=get_dna_color(p(5), pal_type), style=plot.style_area, histbase=0)
plot(2, "L2", color=get_dna_color(p(10), pal_type), style=plot.style_area, histbase=1)
plot(3, "L3", color=get_dna_color(p(15), pal_type), style=plot.style_area, histbase=2)
plot(4, "L4", color=get_dna_color(p(20), pal_type), style=plot.style_area, histbase=3)
plot(5, "L5", color=get_dna_color(p(25), pal_type), style=plot.style_area, histbase=4)
plot(6, "L6", color=get_dna_color(p(30), pal_type), style=plot.style_area, histbase=5)
plot(7, "L7", color=get_dna_color(p(35), pal_type), style=plot.style_area, histbase=6)
plot(8, "L8", color=get_dna_color(p(40), pal_type), style=plot.style_area, histbase=7)
plot(9, "L9", color=get_dna_color(p(45), pal_type), style=plot.style_area, histbase=8)
plot(10, "L10", color=get_dna_color(p(50), pal_type), style=plot.style_area, histbase=9)
plot(11, "L11", color=get_dna_color(p(55), pal_type), style=plot.style_area, histbase=10)
plot(12, "L12", color=get_dna_color(p(60), pal_type), style=plot.style_area, histbase=11)
plot(13, "L13", color=get_dna_color(p(65), pal_type), style=plot.style_area, histbase=12)
plot(14, "L14", color=get_dna_color(p(70), pal_type), style=plot.style_area, histbase=13)
plot(15, "L15", color=get_dna_color(p(75), pal_type), style=plot.style_area, histbase=14)
plot(16, "L16", color=get_dna_color(p(80), pal_type), style=plot.style_area, histbase=15)
plot(17, "L17", color=get_dna_color(p(85), pal_type), style=plot.style_area, histbase=16)
plot(18, "L18", color=get_dna_color(p(90), pal_type), style=plot.style_area, histbase=17)
plot(19, "L19", color=get_dna_color(p(95), pal_type), style=plot.style_area, histbase=18)
plot(20, "L20", color=get_dna_color(p(100), pal_type), style=plot.style_area, histbase=19)

```

```
plot(pi ? p(5) / 5 : na, "Power Index", color=color.white, linewidth=2)
```

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// -----
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```
// Part 5 defining the buy sell blocks
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// -----
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```
var box buyBox = box.new(na, na, na, na, bgcolor=color.new(pal_elite_bull, 20),  
border_color=pal_elite_bull)
```

```
var box sellBox = box.new(na, na, na, na, bgcolor=color.new(pal_elite_bear, 20),  
border_color=pal_elite_bear)
```

```
var label bLabel = label.new(na, na, "", style=label.style_label_down, textcolor=pal_elite_bull,  
color=color.new(color.black, 100))
```

```
var label sLabel = label.new(na, na, "", style=label.style_label_down, textcolor=pal_elite_bear,  
color=color.new(color.black, 100))
```

```
if use_data_mode == "Footprint" and barstate.islast
```

```
float total = fp_buy_vol + fp_sell_vol
```

```
float b_perc = (fp_buy_vol / math.max(total, 1)) * 100
```

```
float s_perc = (fp_sell_vol / math.max(total, 1)) * 100
```

```
// Scale bars to the 0-20 plot range
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```
float b_h = (fp_buy_vol / math.max(fp_buy_vol, fp_sell_vol, 1)) * 20
```

```
float s_h = (fp_sell_vol / math.max(fp_buy_vol, fp_sell_vol, 1)) * 20
```

```
// Wider Buy Bar (Offset 4 to 10)
```

```
box.set_lefttop(buyBox, bar_index + 4, b_h)
```

```
box.set_rightbottom(buyBox, bar_index + 10, 0)
```

```
// Wider Sell Bar (Offset 12 to 18)
```

```
box.set_lefttop(sellBox, bar_index + 12, s_h)
```

```
box.set_rightbottom(sellBox, bar_index + 18, 0)
```

```
// Re-centered Labels
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```
label.set_xy(bLabel, bar_index + 7, b_h + 1.5)
```

```
label.set_text(bLabel, str.format("{0,number,#}% BUY", b_perc))
```

```
label.set_xy(sLabel, bar_index + 15, s_h + 1.5)
```

```
label.set_text(sLabel, str.format("{0,number,#}% SELL", s_perc))
```

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// -----
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// Part 6: plotting the stats table
// -----
var table gainzTable = table.new(position.top_right, 2, 8, bgcolor=color.new(color.black, 10),
border_width=2, border_color=color.new(pal_neon_gold, 50))

if barstate.islast
    color s_color = main_sig == 1 ? pal_elite_bull : pal_elite_bear
    float total = fp_buy_vol + fp_sell_vol

    table.cell(gainzTable, 0, 0, "SYSTEM ANALYSIS", text_color=pal_neon_gold,
text_size=size×small)
    table.cell(gainzTable, 1, 0, "DATA FEED", text_color=pal_neon_gold, text_size=size.small)

    table.cell(gainzTable, 0, 1, "NB Probability", text_color=color.white, text_halign=text.align_left)
    table.cell(gainzTable, 1, 1, str.format("{0,number,}.#}%", main_prob * 100), bgcolor=s_color,
text_color=color×black)

    table.cell(gainzTable, 0, 2, use_data_mode == "Footprint" ? "Delta" : "Rel Volume",
text_color=color×silver, text_halign=text.align_left)
    table.cell(gainzTable, 1, 2, str.tostring(secondary_metric, ".##"), text_color=color.white)

    if use_data_mode == "Footprint"
        table.cell(gainzTable, 0, 3, "Net Buy %", text_color=pal_elite_bull, text_halign=text.align_left)
        table.cell(gainzTable, 1, 3, str.format("{0,number,}.#}%", (fp_buy_vol/total)*100),
text_color=pal_elite_bull)
        table.cell(gainzTable, 0, 4, "Net Sell %", text_color=pal_elite_bear, text_halign=text.align_left)
        table.cell(gainzTable, 1, 4, str.format("{0,number,}.#}%", (fp_sell_vol/total)*100),
text_color=pal_elite_bear)

    table.cell(gainzTable, 0, 7, "SIGNAL", text_color=color.black, bgcolor=color.white)
    table.cell(gainzTable, 1, 7, main_sig == 1 ? "ELITE LONG" : "ELITE SHORT", bgcolor=s_color,
text_color=color.black)

```